Russell Mercer

Biochemistry student preparing for medical school, with research in radiation oncology, surface chemistry, and cryo-electron tomography, and the code and leadership that tie them together. Enjoys learning new things and being challenged in new ways.

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EDUCATION

Brigham Young University · *Provo, UT* Jan 2024 – Apr 2028 (Expected)
B.S. Biochemistry; Minor in Computer Science, with machine learning courses
GPA 3.99 · Science GPA 3.99 · MCAT 519 (96th percentile)

RESEARCH EXPERIENCE

Biochemistry Research Assistant · *Jensen Lab, BYU* Aug 2026 – Present

- Scoped two cryo-electron tomography projects on bacterial and parasite ultrastructure, both approved by the PI and awaiting biosafety review
- Testing whether an AI coding agent can run the cryo-ET processing pipeline with a human approving every step; poster, Utah CryoEM Symposium 2026
- Building an AI agent with Dr. Jensen that interviews cryo-EM authors to collect the tacit knowledge that never reaches print

Clinical Research Assistant · *UT Southwestern Medical Center, Dallas, TX* May 2026 – Present

- Four studies of MR-guided and biology-guided radiotherapy; first author on three manuscripts, one submitted and two in final co-author circulation
- Rebuilt the department's diffusion MRI pipeline into a web application now held by the Radiation Oncology Application Team
- Wrote the C#, Python, and R extraction and figure pipelines behind the studies

Chemistry Research Assistant · *Stowers Lab, BYU* Apr 2024 – May 2026

- Second author, *Langmuir* 2026, on spin-coated metal-organic framework thin films
- Helped build and maintain a flow reactor and a UHV chamber (~10^{-10} Torr, XPS, mass spec)
- Trained two newer students in spin coating and SEM imaging

CLINICAL EXPERIENCE

Medical Assistant · *Health Plus Clinic, Springville, UT* Jul 2025 – Apr 2026

- Assisted a D.O. and a P.A. with patient care, procedures, and documentation

Volunteer · *Utah State Hospital, Provo, UT* May 2025 – Present

- Guide mental health patients through physical therapy; mentored pediatric mental health patients

LEADERSHIP AND SERVICE

Officer · *BYU AI Association* Sep 2026 – Present

- Lead the weekly AI in Research meeting; speak to other colleges across the university and meet with professors and labs about using AI in their work
- Represented the association at BYU's AI for Social Innovation Conference, on the Data Sovereignty and Inclusion working table

Lead Program Director · *BYU Y-Serve, Early Learning Essentials* Aug 2024 – Present

- Lead director since Mar 2025: recruit volunteers, run the leadership team, schedule classroom shifts, and manage the partnership with the Provo preschool center serving low-income families

Pre-Med Peer Mentor · *BYU Pre-Med Association* Oct 2026 – Present

- Mentor pre-med students on coursework, research, and applications

University Peer Mentor · *BYU Office of First Year Experience* Jun 2024 – Feb 2025

- Co-taught two university courses and mentored 45 freshmen

SKILLS

- **Lab:** cryo-EM sample preparation, bacterial culture, SnapGene, MOF synthesis, SEM, UHV and XPS
- **Code:** Python, R, C#, Java, shell, AI coding agents, imaging pipelines
- **Clinical:** chart review, clinical data abstraction, medical assisting, study design

PUBLICATIONS

- Ralph AD, **Mercer RW**, et al. *Langmuir* 2026;42(6):4475–88.
- **First author on three manuscripts:** one submitted (*Int J Radiat Oncol Biol Phys*) and two in final co-author circulation.
- Co-author on two more in final circulation, with others in preparation.

AWARDS

- **SCCR Fellow**, BYU Simmons Center for Cancer Research, 2026
- **Outstanding Analytical Chemist**, 2025
- **Dean's List**, four semesters, 2024 – 2025
- **Undergraduate Research Award**, each semester, Stowers Lab

PRESENTATIONS

- Utah CryoEM Symposium, 2026
- BYU Simmons Center for Cancer Research, 2026
- BYU Medical Society, 2026
- ACS Spring Meeting, Atlanta, 2026
- BYU Student Research Conference, 2025 and 2026

- **Leadership:** program direction, mentoring, cross-disciplinary teams, AI policy discussion